Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________
Topic: Solving Systems by Graphing **Date:** ____________

Foundation (Jalapeno)

Q1. What is the purpose of graphing a system of linear equations?
(A) To find the slope of one line
(B) To identify the intersection point as a solution
(C) To determine the equation of one line
(D) To find the y-intercept of one line
(E) To find the equation of a circle

Q2. Which of the following systems has no solution?
(A) $x + y = 2$ and $x + y = 3$
(B) $x - y = 1$ and $x + y = 2$
(C) $2x + 2y = 4$ and $x + y = 2$
(D) $x + 2y = 3$ and $2x + 4y = 5$
(E) $3x - 2y = 1$ and $x + 2y = 3$

Q3. What does it mean for two lines to be parallel?
(A) They intersect at exactly one point
(B) They intersect at exactly two points
(C) They do not intersect
(D) They are the same line
(E) They have the same y-intercept

Q4. How many solutions does the system $x + y = 2$ and $2x + 2y = 4$ have?
(A) 0
(B) 1
(C) 2
(D) 3
(E) Infinitely many

Q5. What is the equation of the x-axis?
(A) $x = 0$
(B) $y = 0$
(C) $x + y = 0$
(D) $x - y = 0$
(E) $2x + 2y = 0$

Q6. Which of the following points is a solution to the system $x + y = 2$ and $x - y = 0$?
(A) $(0, 0)$
(B) $(1, 1)$
(C) $(2, 0)$
(D) $(0, 2)$
(E) $(1, 0)$

Q7. What does the intersection point of two lines represent?
(A) The solution to one equation
(B) The solution to both equations
(C) The x-intercept of one line
(D) The y-intercept of one line
(E) The slope of one line

Q8. How many equations are needed to form a system of linear equations?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Open Ended Questions (FRQ)

Q9. Graph the system $x + 2y = 4$ and $2x - y = 3$. Identify the intersection point and explain its significance.

Q10. Solve the system $y = 2x - 1$ and $y = -3x + 2$ by graphing. Explain how you found your solution and what it represents.

Chef's Tips

- Ensure students have graph paper and pencils
- Allow students to use calculators for checking
- Remind students to label their axes and intersection points clearly